

Nexivarin 10 mg Tablets — eCTD Submission
Module 3 — Quality

Submission Type	NDA — 505(b)(1)	Sequence Number	0000
Submission Date	2021-03-15	Section	3.2.S.4
Prepared by	PharmaCorp Inc.	Confidentiality	Proprietary and Confidential

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PharmaCorp Inc.
NDA 123456

Nexivarin 10 mg Tablets

Section 3.2.S.4
Drug Substance — Control

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3.2.S.4 Control of Drug Substance

3.2.S.4.1 Specification

Test	Acceptance Criteria	Reference
Description	White to off-white crystalline powder	Visual inspection
Identification A (IR)	Spectrum conforms to reference standard	In-house IM-01 (ICH Q2)
Identification B (HPLC-UV)	Retention time within $\pm 2\%$ of reference	In-house IM-02
Identification C (Chloride)	Positive reaction for chloride ion	USP <191>
Assay (HPLC, anhydrous basis)	98.0% – 102.0%	In-house AM-01 (ICH Q2(R1))
Impurity A (desmethyl impurity)	NMT 0.10%	In-house AM-02
Impurity B (ring-opened product)	NMT 0.05%	In-house AM-02
Impurity C (oxidation product)	NMT 0.10%	In-house AM-02
Any unspecified impurity	NMT 0.10%	In-house AM-02
Total impurities	NMT 0.50%	In-house AM-02
Residual solvents — Ethanol	NMT 5000 ppm	USP <467> / ICH Q3C
Residual solvents — Isopropanol	NMT 5000 ppm	USP <467> / ICH Q3C
Residual solvents — Ethyl acetate	NMT 5000 ppm	USP <467> / ICH Q3C
Residual solvents — n-Heptane	NMT 5000 ppm	USP <467> / ICH Q3C
Water content (Karl Fischer)	NMT 0.5%	USP <921>
Residue on ignition	NMT 0.1%	USP <281>
Heavy metals (ICP-MS)	As, Cd, Hg, Pb each $\leq$ limit per ICH Q3D	ICH Q3D / ICP-MS
Elemental impurities — Pd (catalyst)	NMT 10 ppm	ICP-MS
Particle size (d10/d50/d90)	d10: 5–25 μm / d50: 20–60 μm / d90: NMT 100 μm	Laser diffraction
Polymorphic form	Form I (per reference diffractogram)	XRPD
Microbial enumeration (TAMC)	NMT 1000 cfu/g	USP <61>
Microbial enumeration (TYMC)	NMT 100 cfu/g	USP <61>
Specified organisms	Absent: E. coli, Salmonella, S. aureus	USP <62>

3.2.S.4.4 Batch Analyses

Batch No.	Size (kg)	Date	Assay (%)	Total Imp (%)	Water (%)	Purpose
NXV-DS-001	50	2019-01-15	100.2	0.18	0.31	Clinical (Phase I)
NXV-DS-002	100	2019-06-22	99.8	0.22	0.28	Clinical (Phase II)
NXV-DS-003	500	2020-03-10	100.5	0.15	0.33	Clinical (Phase III)
NXV-DS-004	500	2020-09-14	99.6	0.19	0.29	Clinical (Phase III)
NXV-DS-VAL-001	800	2020-12-01	100.1	0.20	0.30	Validation
NXV-DS-VAL-002	800	2021-01-08	99.9	0.17	0.27	Validation

Batch No.	Size (kg)	Date	Assay (%)	Total Imp (%)	Water (%)	Purpose
NXV-DS-VAL-003	800	2021-02-05	100.3	0.21	0.32	Validation